

Exercise 3.3

1. For $A = \{1, 2, 3, 4\}$ find the following relations in A . State the domain and range of each relation.

$$A \times A = \{1, 2, 3, 4\} \times \{1, 2, 3, 4\}$$

$$A \times A = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3), (4,4)\}$$

(i) $\{(x, y) \mid y = x\}$

Let

$$R_1 = \{(x, y) \mid y = x\}$$

$$R_1 = \{(1,1), (2,2), (3,3), (4,4)\}$$

$$\text{Dom}(R_1) = \{1, 2, 3, 4\}$$

$$\text{Range}(R_1) = \{1, 2, 3, 4\}$$

(ii) $\{(x, y) \mid y + x = 5\}$

Let

$$R_2 = \{(x, y) \mid y + x = 5\}$$

$$R_2 = \{(1,4), (2,3), (3,2), (4,1)\}$$

$$\text{Dom}(R_2) = \{1, 2, 3, 4\}$$

$$\text{Range}(R_2) = \{1, 2, 3, 4\}$$

(iii) $\{(x, y) \mid x + y < 5\}$

Let

$$R_3 = \{(x, y) \mid x + y < 5\}$$

$$R_3 = \{(1,1), (1,2), (1,3), (2,1), (2,2), (3,1)\}$$

$$\text{Dom}(R_3) = \{1, 2, 3\}$$

$$\text{Range}(R_3) = \{1, 2, 3\}$$

(iv) $\{(x, y) \mid x + y > 5\}$

Let

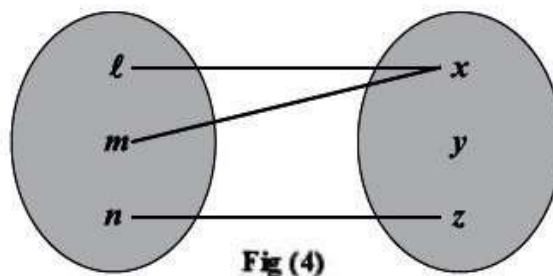
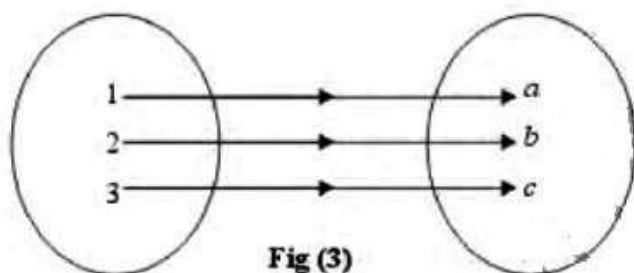
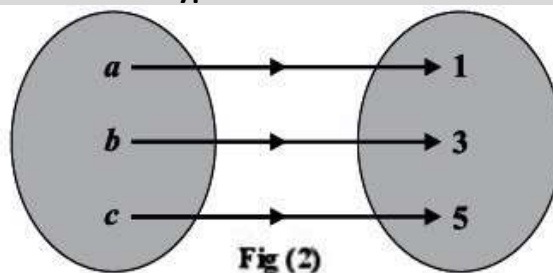
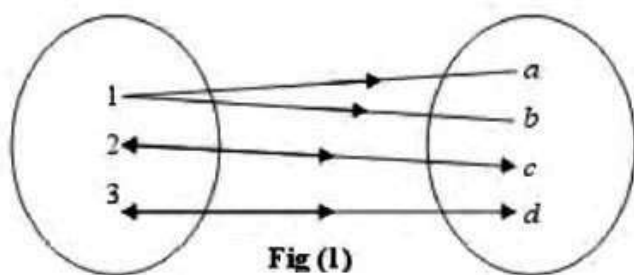
$$R_3 = \{(x, y) \mid x + y > 5\}$$

$$R_3 = \{(2,4), (3,3), (3,4), (4,2), (4,3), (4,4)\}$$

$$\text{Dom}(R_3) = \{2, 3, 4\}$$

$$\text{Range}(R_3) = \{2, 3, 4\}$$

2. Which of the following diagrams represent functions and of which type?



(i) $\{(1, a)(1, b), (2, c)(c, 2), (3, d), (d, 3)\}$ Since domain should not be repeated, but here domain is repeated so it is not a function.

(ii) It is a bijective function (one-one and onto)

(iii) It is a bijective function (one-one and onto)

(iv) It is into function.

3. If $g(x) = 3x + 2$ and $h(x) = x^2 + 1$, then find:

(i) $g(0)$

Since

$$g(x) = 3x + 2$$

Put $x = 0$

$$\begin{aligned} g(0) &= 3(0) + 2 \\ &= 0 + 2 \\ &= 2 \end{aligned}$$

(ii) $g(-3)$

Since

$$g(x) = 3x + 2$$

Put $x = -3$

$$\begin{aligned} g(-3) &= 3(-3) + 2 \\ &= -9 + 2 \\ &= -7 \end{aligned}$$

(iii) $g\left(\frac{2}{3}\right)$

Since

$$g(x) = 3x + 2$$

Put $x = \frac{2}{3}$

$$\begin{aligned} g\left(\frac{2}{3}\right) &= 3\left(\frac{2}{3}\right) + 2 \\ &= 2 + 2 \\ &= 4 \end{aligned}$$

(iv) $h(1)$

Since

$$h(x) = x^2 + 1$$

Put $x = 1$

$$\begin{aligned} h(1) &= (1)^2 + 1 \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

(v) $h(-4)$

Since

$$h(x) = x^2 + 1$$

Put $x = -4$

$$\begin{aligned} h(-4) &= (-4)^2 + 1 \\ &= 16 + 1 \\ &= 17 \end{aligned}$$

(vi) $h\left(-\frac{1}{2}\right)$

Since

$$h(x) = x^2 + 1$$

Put $x = -\frac{1}{2}$

$$\begin{aligned} h\left(-\frac{1}{2}\right) &= \left(-\frac{1}{2}\right)^2 + 1 \\ &= \frac{1}{4} + 1 \\ &= \frac{1+4}{4} \\ &= \frac{5}{4} \end{aligned}$$

4. Given that $f(x) = ax + b + 1$, where a and b are constant numbers. If $f(3) = 8$ and $f(6) = 14$, then find the values of a and b .

Since

$$f(x) = ax + b + 1$$

Put $x = 3$

$$\begin{aligned} f(3) &= a(3) + b + 1 \\ 8 &= 3a + b + 1 \quad \because f(3) = 8 \\ 8 - 1 &= 3a + b \\ 7 &= 3a + b \quad \dots (i) \end{aligned}$$

Now

$$f(x) = ax + b + 1$$

Put $x = 6$

$$\begin{aligned} f(6) &= a(6) + b + 1 \\ 14 &= 6a + b + 1 \quad \because f(6) = 14 \\ 14 - 1 &= 6a + b \\ 13 &= 6a + b \quad \dots (ii) \end{aligned}$$

Subtract equation (i) from equation (ii)

$$\begin{aligned} 13 &= 6a + b \\ \pm 7 &= \pm 3a \pm b \end{aligned}$$

$$6 = 3a$$

$$\frac{6}{3} = a$$

$$2 = a$$

Putting $a = 2$ in equation (i)

$$7 = 3(2) + b$$

$$7 = 6 + b$$

$$7 - 6 = b$$

$$1 = b$$

5. Given that $g(x) = ax + b + 5$, where a and b are constant numbers. If $g(-1) = 0$ and $g(2) = 10$, find the values of a and b .

Since

$$g(x) = ax + b + 5$$

Put $x = -1$

$$g(-1) = a(-1) + b + 5$$

$$0 = -a + b + 5 \quad \because g(-1) = 0$$

$$-5 = -a + b \quad \dots (i)$$

Now

$$g(x) = ax + b + 5$$

Put $x = 2$

$$g(2) = a(2) + b + 5$$

$$10 = 2a + b + 5 \quad \because g(2) = 10$$

$$10 - 5 = 2a + b$$

$$5 = 2a + b \quad \dots (ii)$$

Subtract equation (i) from equation (ii)

$$5 = 2a + b$$

$$\mp 5 = \mp a \pm b$$

$$10 = 3a$$

$$\frac{10}{3} = a$$

Putting $a = \frac{10}{3}$ in equation (i)

$$-5 = -\frac{10}{3} + b$$

$$\frac{10}{3} - 5 = b$$

$$\frac{10 - 15}{3} = b$$

$$\frac{-5}{3} = b$$

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6. Consider the function defined by $f(x) = 5x + 2$. If $f(x) = 32$, find the x value.

Since

$$f(x) = 5x + 2$$

Put $f(x) = 32$

$$32 = 5x + 2$$

$$32 - 2 = 5x$$

$$30 = 5x$$

$$\frac{30}{5} = x$$

$$6 = x$$

$$x = 6$$

7. Consider the function $f(x) = cx^2 + d$, where c and d are constant numbers. If $f(1) = 6$ and $f(-2) = 10$, then find the values of c and d .

Since

$$f(x) = cx^2 + d$$

Put $x = 1$

$$f(1) = c(1)^2 + d$$

$$6 = c(1) + d \quad \because f(1) = 6$$

$$6 = c + d \quad \dots (i)$$

Now

$$f(x) = cx^2 + d$$

Put $x = -2$

$$f(-2) = c(-2)^2 + d$$

$$10 = c(4) + d \quad \because f(-2) = 10$$

$$10 = 4c + d \quad \dots (ii)$$

Subtract equation (i) from equation (ii)

$$10 = 4c + d$$

$$\pm 6 = \pm c \pm d$$

$$4 = 3c$$

$$\frac{4}{3} = c$$

Putting $c = \frac{4}{3}$ in equation (i)

$$6 = \frac{4}{3} + d$$

$$6 - \frac{4}{3} = d$$

$$\frac{18 - 4}{3} = d$$

$$\frac{14}{3} = d$$